



Comprehensive Report on Wet Analysis of EAF, LRF, and VD Slag

Aarti Steel International Limited

Internship Report

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1 Objective

The objectives of this technical study and chemical analysis are multi-fold:

- **Process Comprehension:** To thoroughly understand the metallurgical processes occurring in the Electric Arc Furnace (EAF), Ladle Refining Furnace (LRF), and Vacuum Degassing (VD) stages.
- **Compositional Analysis:** To physically perform wet chemical analysis to calculate the exact percentage of main oxides (SiO_2 , CaO , MgO , MnO , FeO , Fe_2O_3 , and Al_2O_3) present in the slag at each stage.
- **Basicity Determination:** To calculate the basicity (V-ratio) of the slag across different stages and correlate it with the refining capabilities (like desulfurization and dephosphorization) of the heat.
- **Visual Diagnostics:** To understand the physical reasons behind the distinct color changes of the slag (from black to greenish-white to dead white) as it progresses through the steelmaking route.
- **Note:** In the ASIL Chemistry Laboratory, wet chemical analysis is routinely conducted only for Vacuum Degassing (VD) slag. As wet chemical analysis of Electric Arc Furnace (EAF) slag and Ladle Refining Furnace (LRF) slag is not part of the standard laboratory practice, the chemical analyses of these slags were carried out by the myself during this study.

2 Theory and Process Overview

Understanding the origin of the slag requires a deep look into the charging materials, the purpose of each furnace, and the specific flux additions made during the steelmaking process.

2.1 Electric Arc Furnace (EAF)

Purpose and Charging Materials:

- The EAF is the primary melting unit. The charging materials primarily consist of steel scrap and Direct Reduced Iron (DRI/Sponge Iron).
- The main goal here is rapid melting, removal of phosphorus (Dephosphorization), and achieving a baseline carbon and temperature level before tapping.

Flux Additions and Slag Formation:

- **Calcined Lime (CaO):** Added to increase the basicity of the slag. High basicity is required to capture acidic oxides like SiO_2 and P_2O_5 .
- **Calcined Dolomite ($CaO \cdot MgO$):** Added to saturate the slag with MgO . If the slag is starved of MgO , it will chemically attack and erode the MgO -based refractory lining of the furnace.
- **Oxygen Lancing:** Oxygen is blown at high speeds to oxidize impurities. This inevitably oxidizes some iron, creating FeO .

Chemical Reactions in EAF:

- *Silicon Oxidation:* $Si + O_2 \rightarrow SiO_2$
- *Manganese Oxidation:* $Mn + \frac{1}{2}O_2 \rightarrow MnO$
- *Iron Oxidation:* $Fe + \frac{1}{2}O_2 \rightarrow FeO$
- *Dephosphorization:* $2P + 5O_2 + 3CaO \rightarrow 3CaO \cdot P_2O_5$
- *Foamy Slag Creation:* To protect the refractory from the electric arc, carbon is injected to react with FeO . The resulting CO gas bubbles up, foaming the slag.
 $C + FeO \rightarrow Fe + CO \uparrow$

2.2 Ladle Refining Furnace (LRF)

Purpose and Charging Materials:

- The LRF is the secondary metallurgy station. Liquid steel from the EAF is tapped into a ladle and brought here.
- The primary objectives are Deoxidation (killing the steel), Desulfurization, alloying to hit target chemistry, and temperature homogenization via Argon gas purging from the bottom.

Flux Additions and Slag Formation:

- **Deoxidizers:** Aluminum and Silico-Manganese are added. They react with dissolved oxygen to form Al_2O_3 and SiO_2 , which float up into the slag.
- **Synthetic Slag / Lime:** Added to create a highly basic, reducing fluid slag.

Chemical Reactions in LRF:

- *Deoxidation (Aluminum):* $2Al + 3[O] \rightarrow Al_2O_3$
- *Deoxidation (Silicon):* $Si + 2[O] \rightarrow SiO_2$
- *Desulfurization:* Requires a reducing environment (low FeO) and high CaO .
 $[S] + (CaO) \rightarrow (CaS) + [O]$

2.3 Vacuum Degassing (VD)

Purpose and Charging Materials:

- The ladle is placed in a vacuum tank. The extreme drop in pressure forces dissolved gases out of the liquid steel.
- The primary goals are the removal of Hydrogen and Nitrogen, extreme desulfurization, and forcing non-metallic inclusions (like microscopic Al_2O_3 particles) to float up and be absorbed by the slag.

Slag Dynamics in VD:

- No new major fluxes are usually added, but the violent argon stirring under a vacuum ensures aggressive mixing between the steel and the LRF slag.
- The slag here absorbs the final residual inclusions and becomes practically devoid of FeO and MnO .

3 Wet Analysis Procedure

3.1 Sample Preparation

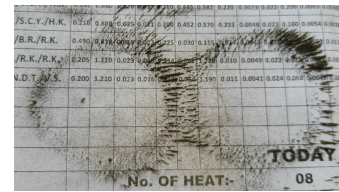
Before any chemical testing can begin, the raw slag sample must be processed into a suitable form for analysis. The slag sample is first crushed using a hammer/jaw crusher to break the large solidified chunks into smaller pieces. These crushed pieces are then screened (sieved) repeatedly to obtain a fine, uniform powder of the slag. During crushing, small uncrushed metallic iron particles (entrapped metal droplets) often get mixed into the slag powder. To remove these, a magnet is passed over/through the powdered sample to separate out these uncrushed iron particles, ensuring that only the pure slag powder remains. Once this magnetic separation is complete, the final fine slag powder is ready to be taken for the wet chemical analysis.



(1) Crushing



(2) Screening



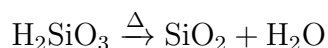
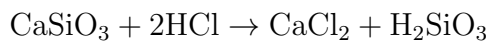
(3) Magnetic Separation

Figure 1: Sample Preparation

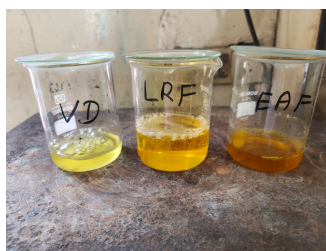
3.2 Preparation and Silica (SiO_2) Determination

- **Weighing:** Accurately weigh 0.5 gm of the finely powdered slag sample.
- **Dissolution:** Transfer to a beaker. Add 50 ml of Hydrochloric Acid (HCl) and 50 ml of distilled water to dissolve the powder. Add 2–3 drops of Nitric Acid (HNO_3) to ensure complete oxidation.
- **Baking:** Heat the mixture on a burner until it is completely dry.
- **Redissolving:** Add acid again and boil. This process turns silica into an insoluble gelatinous form.
- **Filtration:** Filter the solution through an ashless filter paper. The filtrate (liquid) is collected and retained for the R_2O_3 separation step. The residue remaining on the filter paper is the silica.
- **Ignition:** Place the filter paper with the residue into a crucible. Ignite in a muffle furnace at $750^\circ C$ for 30 minutes.

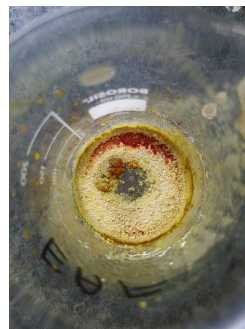
- **Reactions:**



(4) Weighing Sample



(5) Dissolving into HCl



(6) Dried Sample



(7) Filtration of SiO_2



(8) Crucible in Muffle Furnace at 750°C



(9) SiO_2 Final

Figure 2: SiO_2 Determination

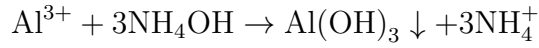
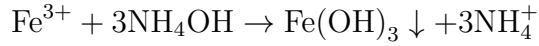
3.3 Separation of R_2O_3 (Iron & Aluminum)

Note: R_2O_3 refers to the combined group of trivalent metal oxides present in the slag, i.e., $\text{Fe}_2\text{O}_3 + \text{Al}_2\text{O}_3 + \text{Cr}_2\text{O}_3 + \dots$, all of which precipitate together under ammoniacal conditions.

- **Starting Material:** The filtrate collected from the SiO_2 step (containing dissolved Fe, Al, Ca, and Mg ions) is used directly for this step.
- **Precipitation:** Add Ammonium Chloride (NH_4Cl) to the filtrate and boil.
- **Ammoniacal Stage:** Slowly add Ammonia (NH_3) solution until the solution becomes ammoniacal (basic). Let it cool.
- **Observation:** Iron precipitates as a fluffy rust-brown gelatinous mass ($\text{Fe}(\text{OH})_3$), and Aluminum precipitates as a gelatinous white mass ($\text{Al}(\text{OH})_3$).

- **Filtration:** Filter the precipitate. The R_2O_3 solids are retained on the filter paper and ignited in a muffle furnace to determine their weight. The new filtrate, which contains Ca and Mg ions, is collected for the next step.

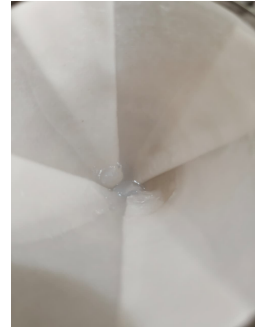
- **Reactions:**



(10) Ammoniacal Solution



(11) R_2O_3 Filtrate of EAF



(12) R_2O_3 of LRF



(13) R_2O_3 Final

Figure 3: R_2O_3 Determination

3.4 Titration for Calcium (CaO) and Magnesium (MgO)

- **Solution Preparation:** The filtrate from the R_2O_3 separation (containing Ca^{2+} and Mg^{2+} ions) is transferred to a volumetric flask and made up to **500 ml** with distilled water. For each titration, **10 ml** of this solution is pipetted out, so the effective slag sample represented in each aliquot is:

$$\text{Effective Sample Weight} = 0.5 \times \frac{10}{500} = 0.01 \text{ gm}$$

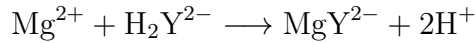
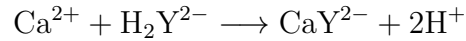
- **Total Hardness ($Ca + Mg$):** The 10 ml aliquot is titrated with standard EDTA (H_2Y^{2-}) solution to find the combined Ca and Mg content. Solochrome Black T

is used as the indicator. The titration color shifts sharply from wine red to a deep blue endpoint.

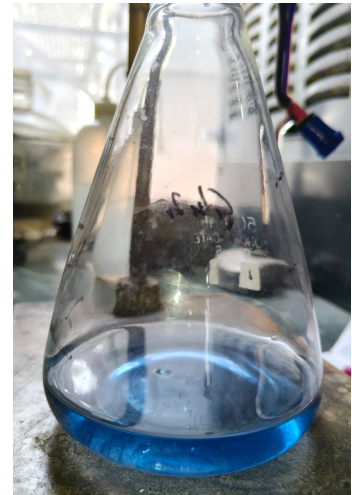
- **Calcium Specific (CaO):** A fresh 10 ml aliquot is adjusted to a high pH using NaOH. Add 2–3 drops of PNDR indicator. Titrate with EDTA. The titration color shifts sharply from wine red to a deep blue endpoint.
- **Magnesium Specific (MgO):** The MgO reading is found by subtracting the CaO titration reading from the Total Hardness reading:

$$MgO \text{ reading (cc)} = \text{Total Hardness reading (cc)} - CaO \text{ reading (cc)}$$

- **Reactions** (where H_2Y^{2-} represents the EDTA compound):



(14) Wine Red Solution Before Titration



(15) Sharp Blue Color After Titration

Figure 4: CaO and MgO Test

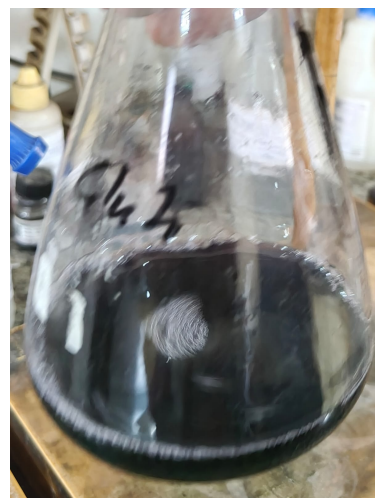
3.5 Titration for Iron Oxide (FeO)

- **Sample Prep:** Weigh a fresh 0.5 gm slag sample. Dissolve in HCl.
- **Conditioning for EAF Slag:** After dissolution in HCl, add **Sodium Bicarbonate** ($NaHCO_3$) to create a slightly alkaline, protective atmosphere to prevent air oxidation of Fe^{2+} . Then add Orthophosphoric Acid (H_3PO_4) and **Mercuric Chloride** ($HgCl_2$) to remove any interfering Fe^{3+} by reduction, and Barium Diphenylamine Sulfonate (BDAS) as the indicator.

- **Conditioning for LRF and VD Slag:** After dissolution in HCl, add **Zinc metal (Zn)** to reduce any Fe^{3+} back to Fe^{2+} . Then add Orthophosphoric Acid (H_3PO_4) and Barium Diphenylamine Sulfonate (BDAS) as the indicator.
- **Titration:** Titrate against a standard Potassium Dichromate ($K_2Cr_2O_7$) solution (Normality = N/10).
- **Endpoint:** The color shifts from pale green to a sharp violet-blue.
- **Reaction:** $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$



(18) Pale Green Color Before Titration



(19) Violet-Blue Color After Titration

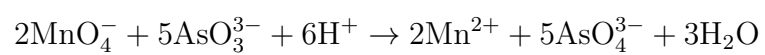
Figure 5: FeO Test

3.6 Titration for Manganese Oxide (MnO)

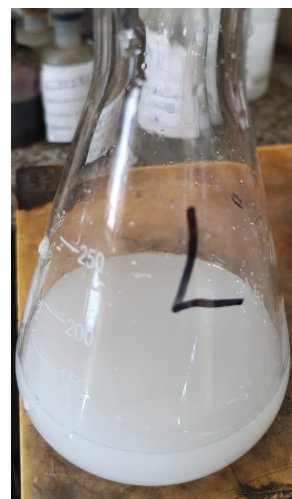
- **Sample Prep:** Weigh exactly 0.2 gm of the slag sample. Dissolve in 20 ml of HNO_3 and 80 ml of water.
- **Oxidation:** Add Silver Nitrate ($AgNO_3$) as a catalyst, and Ammonium Persulfate ($(NH_4)_2S_2O_8$) as the oxidizer. Boil the solution. It will turn a deep pink/purple color indicating the formation of permanganate ions.
- **Indicator:** Add **Sodium Chloride ($NaCl$)** as indicator to the solution before titration.
- **Titration:** Titrate against standard Sodium Arsenite ($NaAsO_2$) solution (Normality = 0.0154 N) until the deep purple color turns completely colorless.
- **Oxidation Reaction:**

$$2Mn^{2+} + 5S_2O_8^{2-} + 8H_2O \xrightarrow{Ag^+} 2MnO_4^- + 10SO_4^{2-} + 16H^+$$

- **Titration Reaction:**



(16) Pink Color Before Titration



(17) Colorless Solution After Titration

Figure 6: *MnO* Test

4 Observations and Calculations

4.1 Burette / Weight Readings

The raw readings obtained from the wet analysis for both steel grades are tabulated below. The values for SiO_2 and R_2O_3 are weights in grams, while the values for CaO , Total Hardness, MnO , and FeO are volumes in cc (ml).

Table 1: Raw Wet Analysis Readings for Heat EN353

Parameter	Unit	EAF	LRF	VD
SiO_2	gm	0.082	0.041	0.046
R_2O_3	gm	0.150	0.137	0.144
CaO	cc	7.30	10.20	9.00
Total Hardness (Ca+Mg)	cc	10.00	11.50	11.00
MnO	cc	19.50	1.50	3.50
FeO	cc	8.00	0.50	0.80

Table 2: Raw Wet Analysis Readings for Heat 40CrMo4

Parameter	Unit	EAF	LRF	VD
SiO_2	gm	0.099	0.051	0.056
R_2O_3	gm	0.130	0.128	0.160
CaO	cc	6.80	9.90	9.10
Total Hardness (Ca+Mg)	cc	10.10	11.30	10.90
MnO	cc	16.10	1.20	4.00
FeO	cc	6.10	0.30	0.30

4.2 Formula Breakdown and Calculation Method

To convert the burette readings (in cc or ml) into mass percentages, we rely on the principles of volumetric equivalence.

4.2.1 Basicity Formula

Basicity (V-ratio) is the ratio of the main basic oxide to the main acidic oxide in the slag. It is the primary metric for determining the slag's ability to hold sulfur and phosphorus.

$$\text{Basicity (V)} = \frac{\%CaO}{\%SiO_2}$$

4.2.2 Calculation for Silica (SiO_2)

Silica is determined gravimetrically (by weight), not by titration.

$$\%SiO_2 = \left(\frac{\text{Weight of ignited residue}}{\text{Sample Weight (0.5 g)}} \right) \times 100$$

Therefore: $\%SiO_2 = \text{Reading (Weight)} \times 200$

4.2.3 Calculation for R_2O_3

R_2O_3 is also determined gravimetrically, identical in method to silica:

$$\%R_2O_3 = \left(\frac{\text{Weight of ignited precipitate}}{\text{Sample Weight (0.5 g)}} \right) \times 100$$

Therefore: $\%R_2O_3 = \text{Reading (Weight)} \times 200$

4.2.4 Calculation for Calcium Oxide (CaO)

The general formula for finding the percentage of an oxide via titration is:

$$\%Oxide = \frac{\text{Reading} \times \text{Normality} \times \text{Molar Mass}}{n_{factor} \times \text{Effective Sample Weight} \times 10}$$

Since the filtrate from the R_2O_3 step is made up to 500 ml and only 10 ml is taken for titration, the effective sample weight in the aliquot is:

$$\text{Effective Sample Weight} = 0.5 \times \frac{10}{500} = 0.01 \text{ gm}$$

For Calcium, the valency (n_{factor}) is 2. The molar mass of CaO is 56.08 g/mol. The normality of the EDTA used is 1/50 (0.02 N).

$$\%CaO = \frac{\text{Reading} \times \frac{1}{50} \times 56.08}{2 \times 0.01 \times 10} = \frac{\text{Reading} \times 0.02 \times 56.08}{0.2}$$

Simplified Constant: $\%CaO = \text{Reading (cc)} \times 5.608$

4.2.5 Calculation for Magnesium Oxide (MgO)

Using the same principles, the molar mass of MgO is 40.30 g/mol, and $n_{factor} = 2$. The effective sample weight is again 0.01 gm (same 500 ml dilution, 10 ml aliquot).

$$\%MgO = \frac{(\text{Total Hardness Reading} - \text{CaO Reading}) \times \frac{1}{50} \times 40.30}{2 \times 0.01 \times 10}$$

Simplified Constant: $\%MgO = (\text{Total Hardness Reading} - \text{CaO Reading}) \times 4.03$

4.2.6 Calculation for Iron Oxide (FeO)

The normality of the Potassium Dichromate used is N/10 (0.1 N). The molar mass of FeO is 71.84 g/mol. Iron transitions from Fe^{2+} to Fe^{3+} , so the $n_{factor} = 1$. The sample weight for FeO titration is 0.5 gm directly (no dilution).

$$\%FeO = \frac{\text{Reading} \times \frac{1}{10} \times 71.84}{1 \times 0.5 \times 10}$$

Simplified Constant: $\%FeO = \text{Reading (cc)} \times 1.4368$

4.2.7 Calculation for Manganese Oxide (MnO)

Understanding Normality of Sodium Arsenite ($NaAsO_2$):

- Molecular Weight of $NaAsO_2 = 23 (Na) + 75 (As) + 32 (O_2) = 130$ g/mol.
- During titration, Arsenic oxidizes from +3 to +5, so its $n_{factor} = 2$.
- Equivalent weight = $130/2 = 65$.
- If 1 gm of Sodium Arsenite is dissolved per Liter, the Normality $N = \frac{1}{65} = 0.0154$ N.

Understanding the Mn n -factor: During the titration, the purple permanganate ion (MnO_4^-) where Mn is in the +7 state is reduced back to the colorless Mn^{2+} state. The change in oxidation state (n_{factor}) is **5**. The molar mass of MnO is 70.94 g/mol. Sample weight is 0.2 g.

$$\%MnO = \frac{\text{Reading} \times 0.0154 \times 70.94}{5 \times 0.2 \times 10}$$

Simplified Constant: $\%MnO = \text{Reading (cc)} \times 0.1092$

4.2.8 Calculation for Alumina (Al_2O_3)

Alumina is calculated by subtracting the iron content from the total R_2O_3 precipitate weight. First, convert FeO to Fe_2O_3 :

$$\%Fe_2O_3 = 1.11 \times \%FeO$$

Then subtract from the total R_2O_3 :

$$\%Al_2O_3 = \%R_2O_3 - \%Fe_2O_3$$

5 Results

The wet analysis was carried out on slag samples collected from two different steel grades: **EN353** and **40CrMo4**. The calculated percentages for each phase of the steelmaking process are documented below.

5.1 Results for Steel Grade: EN353

Table 3: Slag Composition Percentage for Heat EN353

Component	EAF (%)	LRF (%)	VD (%)
SiO ₂	16.40	8.20	9.20
R ₂ O ₃	30.00	27.40	28.80
CaO	40.94	57.20	50.47
MgO	10.88	5.24	8.06
MnO	2.13	0.16	0.38
FeO	11.49	0.72	1.15
Fe ₂ O ₃	12.76	0.80	1.28
Al ₂ O ₃	17.34	26.60	27.52

5.2 Results for Steel Grade: 40CrMo4

Table 4: Slag Composition Percentage for Heat 40CrMo4

Component	EAF (%)	LRF (%)	VD (%)
SiO ₂	19.80	10.20	11.20
R ₂ O ₃	26.00	25.60	32.00
CaO	38.13	55.52	51.03
MgO	13.30	5.64	7.25
MnO	1.76	0.13	0.44
FeO	8.76	0.43	0.43
Fe ₂ O ₃	9.72	0.48	0.48
Al ₂ O ₃	16.28	25.12	31.52

5.3 Basicity Comparison

The basicity ($V = \frac{CaO}{SiO_2}$) defines the metallurgical capability of the slag.

Table 5: Basicity (V-Ratio) Comparison across Grades

Steel Grade	EAF Basicity	LRF Basicity	VD Basicity
EN353	2.50	6.98	5.48
40CrMo4	1.93	5.44	4.56

6 Comparison with Standard Values

To ensure the process is under metallurgical control, the observed results are cross-referenced against standard values from the MSTS/NST Book (for EAF) and Ahindra Ghosh (for LRF and VD). Values highlighted in **bold** fall outside the standard range.

6.1 Comparison for Steel Grade: EN353

Table 6: Standard Range vs. Experimental Observations – Heat EN353

Stage	Component	Standard Range	Observed (EN353)
EAF	SiO ₂	12.0% – 20.0%	16.40%
	CaO	35.0% – 45.0%	40.94%
	MgO	8.0% – 12.0%	10.88%
	MnO	2.0% – 5.0%	2.13%
	Al ₂ O ₃	3.0% – 10.0%	17.34%
	FeO	15.0% – 30.0%	11.49%
	Basicity	1.8 – 2.5	2.50
LRF	SiO ₂	5.0% – 10.0%	8.20%
	CaO	50.0% – 60.0%	57.20%
	MgO	5.0% – 8.0%	5.24%
	Al ₂ O ₃	20.0% – 30.0%	26.60%
	FeO + MnO	< 1.0%	0.88%
	Basicity	4.0 – 6.0	6.98
VD	SiO ₂	4.0% – 8.0%	9.20%
	CaO	55.0% – 62.0%	50.47%
	MgO	6.0% – 8.0%	8.06%
	Al ₂ O ₃	25.0% – 32.0%	27.52%
	FeO + MnO	< 0.5%	1.53%
	Basicity	5.0 – 8.0	5.48

6.2 Comparison for Steel Grade: 40CrMo4

Table 7: Standard Range vs. Experimental Observations – Heat 40CrMo4

Stage	Component	Standard Range	Observed (40CrMo4)
EAF	SiO ₂	12.0% – 20.0%	19.80%
	CaO	35.0% – 45.0%	38.13%
	MgO	8.0% – 12.0%	13.30%
	MnO	2.0% – 5.0%	1.76%
	Al ₂ O ₃	3.0% – 10.0%	16.28%
	FeO	15.0% – 30.0%	8.76%
	Basicity	1.8 – 2.5	1.93
LRF	SiO ₂	5.0% – 10.0%	10.20%
	CaO	50.0% – 60.0%	55.52%
	MgO	5.0% – 8.0%	5.64%
	Al ₂ O ₃	20.0% – 30.0%	25.12%
	FeO + MnO	< 1.0%	0.56%
	Basicity	4.0 – 6.0	5.44
VD	SiO ₂	4.0% – 8.0%	11.20%
	CaO	55.0% – 62.0%	51.03%
	MgO	6.0% – 8.0%	7.25%
	Al ₂ O ₃	25.0% – 32.0%	31.52%
	FeO + MnO	< 0.5%	0.87%
	Basicity	5.0 – 8.0	4.56

Note on EAF Alumina Content: The Al₂O₃ percentage observed in the EAF slag (17.34% for EN353 and 16.28% for 40CrMo4) appears elevated compared to the standard range of 3.0%–10.0%. This is because the EAF slag sample is collected from the same spoon used to take the molten steel sample for spectroscopic analysis. Prior to sampling, a 4 mm aluminum wire is added to the spoon to reduce porosity in the sample disc, as the steel at this stage has a relatively high dissolved oxygen content. This additional aluminum oxidizes and contributes extra Al₂O₃ to the slag adhering to the spoon, resulting in a higher measured alumina percentage than is present in the bulk furnace slag.

Note on EAF FeO Content: The FeO levels measured in both heats (11.49% for EN353 and 8.76% for 40CrMo4) fall below the standard range of 15.0%–30.0%. The

lower FeO in the 40CrMo4 heat in particular indicates tighter oxygen control during EAF operation for this grade.

Note on LRF Basicity for EN353: The LRF basicity of 6.98 for EN353 slightly exceeds the standard guideline range of 4.0–6.0. This reflects aggressive synthetic lime addition to create a highly desulfurizing environment for this heat, and is not detrimental to the process.

Note on VD Deviations: The slight deviations in VD CaO, SiO_2 , and FeO+MnO compared to ideal standards are typical consequences of active inclusion absorption during deep vacuum treatment, which dilutes the slag composition.

General Observation: In both heats, the basicity jumps massively from the EAF to the LRF stage due to the heavy addition of synthetic lime to create a desulfurizing environment.

7 Conclusions and Slag Color Diagnostics

Based on the quantitative wet analysis and visual observations of the process, several key metallurgical conclusions can be drawn:

7.1 Understanding Slag Color Variations

One of the most defining physical characteristics of steelmaking slag is its color, which shifts drastically due to the oxidation state of transition metals (primarily Iron and Manganese) within it.

- **EAF Slag (Black / Dark Brown):** The EAF slag was observed to be pitch black. The chemical analysis proves this is due to the highly oxidizing environment of the furnace. The EAF slag contains high amounts of FeO (up to 11.49%) and MnO (up to 2.13%). These transition metal oxides act as strong pigments, completely darkening the slag.



Figure 7: EAF Slag (Black Color)

- **LRF Slag (Greenish-White / Grey):** As the liquid steel moves to the LRF, powerful deoxidizers (Aluminum and Silicon) are added. The analysis shows FeO dropping drastically below 1% (e.g., 0.43% to 0.72%). As the heavy metal oxides are reduced back into the steel bath, the slag loses its dark color. The remaining slag is mostly basic calcium-aluminate. Traces of sulfur forming CaS , combined with microscopic remnants of reduced Mn/Cr ions, give this crumbling basic slag a distinct pale green or greyish-white appearance.



Figure 8: LRF Slag (Greenish-White Color)

- **VD Slag (Dead White / Pure White):** After the intense vacuum treatment, the slag becomes “dead”, meaning it is completely reduced. The analysis shows the absolute minimum levels of FeO and MnO . The resulting matrix is almost entirely pure $CaO-Al_2O_3-SiO_2$ (Calcium Aluminate and Calcium Silicate). Because there are no colored transition metal oxides left to pigment the matrix, the slag cools into a pure, powdery dead-white dust. The powdery nature is caused by the phase transformation of dicalcium silicate upon cooling.



Figure 9: VD Slag (Dead White Color)

7.2 Process Efficiency

- **Refractory Protection:** The MgO levels across all stages remained between 5% and 13%. This indicates that dolomite charging practices are effective, ensuring the slag is saturated and does not erode the furnace linings.

- **Desulfurization Capacity:** The massive increase in Basicity (V-ratio) from ≈ 2.0 in the EAF to > 5.0 in the LRF confirms that the synthetic slag additions are working effectively. A high basicity combined with low FeO creates the exact thermodynamic conditions required to pull sulfur out of the molten steel.
- **Inclusion Flotation:** The high alumina (Al_2O_3) content in the LRF and VD stages (ranging from 25% to 31%) proves that the aluminum wire injections are successfully stripping dissolved oxygen from the steel bath. The basic slag effectively traps these resulting alumina inclusions, guaranteeing clean, high-grade steel.